

1 The original primitive source and its adjacent Gamma metric

This calculation continues the complete LCA15–21 polynomial quotient construction. Its new conclusions are the explicit codimension-three truncation, its exact rank-one native metric change, and a finite coefficient-product bound for that change. All spaces and measures below are the original ones. The connecting target metric and its later long return remain the literal target metrics; the source computation here is at the two specified low cutoffs.

1.1 Original objects and the polynomial quotient map

Fix the actual simple quartet, its invertible period matrix H , logarithm branch, and full conductor coefficients. Retain

$$\begin{aligned} \beta_* &= \frac{1}{2} - \delta - i\gamma, & \beta_{n,a,b} &= n\beta_* + 2a\delta + 2bi\gamma, & \delta, \gamma &> 0, \\ E_A(z) &= \sum_{a,b=0}^8 a_{ab} e^{\beta_{8,a,b}z}, & \mu_r &= E_A^{(r)}(0), & v = \text{ord}_0 E_A \leq 80, & \mu_v \neq 0, \\ L_i(Z, W) &= (H(ZW, Z, W, 1)^T)_i, & b_{i,a,b} &= [Z^a W^b] L_i^4, & B_i(z) &= \sum_{a,b=0}^4 b_{i,a,b} e^{\beta_{4,a,b}z}, \\ b_i &= B_i(0) = \sum_{a,b=0}^4 b_{i,a,b}, & \sum_{i=1}^4 D_i(z) B_i(z) &= 1. \end{aligned} \tag{PDS1}$$

The D_i are the proved original entire Bezout expressions. In particular at least one b_i is nonzero. Keep the full biform conductor, including repeated components and zero edge coefficients. Set

$$\begin{aligned} k &= 4l + 1 \geq 9, & j &= k + 4, & q_k &= (k + 1)^2, & Q &= (k + 5)^2, \\ e &\in \{0, 1\}, & D_e &= Q - v - 1 + e, & \ell_e &= 8k + 23 + e, & d_e &= \ell_e + 1, \\ \chi_n(S) &= \prod_{a,b=0}^n (S - \beta_{n,a,b}), & (\mathcal{T}_{AP})(S') &= \sum_{a,b=0}^8 a_{ab} p(S' + \beta_{8,a,b}), \\ (\mathcal{T}_{B_i p})(S) &= \sum_{a,b=0}^4 b_{i,a,b} p(S + \beta_{4,a,b}). \end{aligned} \tag{PDS2}$$

All dual functionals are complex-linear. For a finite germ F through degree D_e , let $\Lambda_F(S'^r) = F^{(r)}(0)$. Let $\mathcal{H}_e = \mathbb{C}[S']_{\leq D_e}^\vee$, and let $\mathcal{S}_e \subset \mathcal{H}_e$ be the jets of the original analytic germs

$$F(z) = \frac{N(z)}{E_A(z)}, \quad N(z) = \sum_{a,b=0}^k n_{ab} e^{\beta_{k,a,b}z}, \quad N^{(r)}(0) = 0 \quad (0 \leq r < v). \tag{PDS3}$$

Define the unchanged relation polynomials and quotient coordinates by

$$r_a(S') = \mathcal{T}_A(\chi_k(S) S^a), \quad 0 \leq a \leq \ell_e, \quad \pi_e(F) = (\Lambda_F(r_a))_{a=0}^{\ell_e}. \tag{PDS4}$$

Then π_e is onto and $\ker \pi_e = \mathcal{S}_e$. Here is the complete kernel argument. The distinct upper k nodes give a Vandermonde matrix, whose first v rows have rank v since $v \leq 80 < q_k$. The numerator space in PDS3 therefore has dimension $q_k - v$. Its jet map is injective: if F vanishes through degree D_e , then $N = E_A F$ vanishes through degree $D_e + v \geq q_k - 1$; the first q_k Vandermonde rows force every $n_{ab} = 0$. The Leibniz identity gives $\Lambda_F(\mathcal{T}_{AP}) = \Lambda_{E_A F}(p)$, so all r_a annihilate PDS3. Finally

$$\mathcal{T}_A S^m = \sum_{r=0}^m \binom{m}{r} \mu_r S'^{m-r}, \quad \deg r_a = q_k + a - v, \quad \text{lc}(r_a) = \mu_v \binom{q_k + a}{v}. \tag{PDS5}$$

These successive degrees prove independence. Their common kernel has dimension $D_e + 1 - d_e = q_k - v$, proving the assertion. This is precisely the LCA16 quotient, with no new row norm. Every original lower degree- $(k - 8)$ exponential is in $\mathcal{S}_e$, since its product with E_A has an allowed degree- k numerator. Thus taking the original lower-gauge quotient first and then quotienting by its analytic-source image gives exactly the same affine fibres as $\mathcal{H}_e/\mathcal{S}_e$.

1.2 The full original Gamma dual metric

The source member of the two original order pairs is $s = 1$ or $s = k$, and its centre is $c = k/2 - 4$. Keep its entire measure and recurrence:

$$M_s = (2\pi)^{s/2}, \quad dm_s(y) = \frac{M_s 2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy, \quad h_{n,s} = M_s n! (s/2)_n, \\ p_{0,s} = 1, \quad p_{1,s} = y, \quad p_{n+1,s} = yp_{n,s} - n(n-1 + s/2)p_{n-1,s}, \quad \phi_n(S') = p_{n,s}((S' - c)/i)/\sqrt{h_{n,s}}. \quad (\text{PDS6})$$

These are the original orthonormal polynomials for the norm $\|p\|_{s,c}^2 = \int_{\mathbb{R}} |p(c + iy)|^2 dm_s(y)$, as established in the original Gamma construction reproduced in the cumulative providers. The coefficient isometry uses this complete measure, rather than dividing its mass out. Let B_e be the triangular matrix defined by $S'^r = \sum_{n=0}^{D_e} (B_e)_{nr} \phi_n(S')$ and let

$$C_e = (B_e \text{coeff}_{S'} r_a)_{a=0}^{\ell_e}, \quad H_e = C_e^* C_e > 0, \quad G_e = \overline{H_e^{-1}}. \quad (\text{PDS7})$$

The columns of C_e are polynomial coefficients in the original orthonormal frame. For $w = (\Lambda_F(\phi_n))_n$ the literal quotient row map is $x = C_e^\top w$. Its minimum-norm lift is $w = \overline{C_e} \overline{H_e}^{-1} x$ and has squared norm $x^* G_e x$. Indeed $C_e^\top \overline{C_e} = \overline{H_e}$, and the displayed lift is orthogonal to $\ker C_e^\top$. This proves both the full complex-linear dual orientation and the quotient metric, including every conjugation in PDS7. The same metric results from the two successive minima over the original lower gauges and analytic source, by the equality of fibres proved after PDS5.

The degree-one primitive condition at the first endpoint is empty. At the second endpoint its four original rows are

$$R_i(S) = \frac{\mathcal{T}_{B_i} \chi_j(S)}{\chi_k(S)}, \quad W = (\text{coeff}_S R_i)_{i=1}^4 \in \mathbb{C}^{d_1 \times 4}, \quad \mathcal{P}_0 = \mathbb{C}^{d_0}, \quad \mathcal{P}_1 = \ker W^\top \subset \mathbb{C}^{d_1}. \quad (\text{PDS8})$$

Here $\mathcal{P}_e$ denotes the primitive quotient source, not a biform section space. All degree- k roots become degree- j roots after every degree-four shift; hence the numerator is divisible by χ_k and $\deg R_i \leq \ell_1$. Translation sums commute, so $\mathcal{T}_A(\chi_k R_i) = \mathcal{T}_{B_i} \mathcal{T}_A \chi_j$. This is exactly the original positive degree-zero exterior differential, and its row is $W^\top$, as in LCA18–19. The full independence proof LCA20–21 applies to the actual invertible period, without a genericity exclusion: the four L_i^4 are independent on the rank-four Segre quadric, and the 25 distinct translations of the degree- $Q - v \geq 116$ polynomial $\mathcal{T}_A \chi_j$ give an injective Vandermonde moment map. Thus $\text{rank } W = 4$. The complete LCA20–21 proof is retained as a provider in the cumulative source, rather than replacing this fact by a new period assumption.

The metric on $\mathcal{P}_e$ is the restriction of G_e . At $e = 1$ its exact polynomial dual is

$$\text{span}(r_0, \dots, r_{\ell_1}) / \text{span}(\mathcal{T}_A(\chi_k R_i))_{i=1}^4.$$

To verify the metric as well as the vector-space statement, choose any fixed full basis matrix Z_1 of $\ker W^\top$. A coefficient vector f of a relation polynomial restricts to the functional $Z_1^\top f$ on this basis. Minimizing $f^* H_1 f$ over all lifts with $Z_1^\top f = a$ gives

$$\min_{Z_1^\top f = a} f^* H_1 f = a^* (Z_1^\top H_1^{-1} \overline{Z_1})^{-1} a = a^* (\overline{Z_1^* G_1 Z_1})^{-1} a. \quad (\text{PDS9})$$

The minimizer follows by the adjoint of $Z_1^\top H_1^{-1/2}$; subtracting it leaves an orthogonal kernel vector. The kernel of $Z_1^\top$ is exactly $\text{im } W$, by dimensions and $W^\top Z_1 = 0$. This proves the stated original quotient metric and its exact dual morphism, with no deletion of the four row polynomials.

1.3 The exact codimension-three truncation

The coefficient of S^{ℓ_1} in R_i is b_i , since both χ_j and χ_k are monic. Choose a fixed original index i_* with $b_{i_*} \neq 0$, write $b_* = b_{i_*}$, and retain

$$R_{i_*}(S) = \sum_{a=0}^{\ell_0} (u_*)_a S^a + b_* S^{\ell_1}, \quad C_i(S) = R_i(S) - \frac{b_i}{b_*} R_{i_*}(S) \quad (i \neq i_*). \quad (\text{PDS10})$$

These displayed combinations are exact coordinates for the intersection of the original four-row span with lower degree; the original rows and coefficients in PDS8 remain fixed. The three C_i have degree at most ℓ_0 and are independent: a relation among them would give a relation among all four R_i , whose coefficients on indices other than i_* are unchanged. Let V have the three coefficient columns C_i , in the original index order with i_* omitted. Then

$$\mathcal{Z} = \ker V^T \subset \mathbb{C}^{d_0}, \quad \dim \mathcal{Z} = d_0 - 3 = 8k + 21, \quad E = \begin{bmatrix} I_{d_0} \\ -u_*^T/b_* \end{bmatrix}, \quad (\text{PDS11})$$

$$\text{pr}_0 : \mathcal{P}_1 \xrightarrow{\simeq} \mathcal{Z}, \quad \text{pr}_0^{-1} = E|_{\mathcal{Z}}.$$

In fact the i_* row of $W^T(x_0, x_{\text{top}}) = 0$ gives $x_{\text{top}} = -u_*^T x_0 / b_*$. Substitution in the other three rows gives precisely $V^T x_0 = 0$. The first d_0 relation polynomials in PDS4 are unchanged at the two cutoffs, so pr_0 is also the actual jet truncation on the primitive quotient source. If an upper analytic source is added its truncated jet is an old analytic source; hence this map is well-defined on the original quotient. Surjectivity and the displayed inverse follow from the equations just proved.

1.4 One rank-one update, with every frame retained

Choose an arbitrary fixed full column basis Z of $\mathcal{Z}$; use EZ at the upper endpoint. For the same source order s and centre c , the old orthonormal polynomials are unchanged when the cutoff increases. In particular the old C_0 is padded by a zero last row in C_1 . Write the exact block matrix and Schur scalar

$$H_1 = \begin{bmatrix} H_0 & b \\ b^* & a \end{bmatrix}, \quad b = C_0^* c_{\text{top}}, \quad a = c_{\text{top}}^* c_{\text{top}}, \quad \sigma = a - b^* H_0^{-1} b > 0, \quad (\text{PDS12})$$

where c_{top} is the full coefficient column of r_{ℓ_1} . Padded columns are understood in the products. Define

$$\eta = -u_*^T/b_* - b^T \overline{H}_0^{-1}, \quad K_0 = Z^* \overline{H}_0^{-1} Z, \quad K_1 = (EZ)^* \overline{H}_1^{-1} (EZ). \quad (\text{PDS13})$$

Then the complete metric comparison is

$$\boxed{K_1 = K_0 + \frac{(\eta Z)^*(\eta Z)}{\sigma}, \quad \frac{\det K_1}{\det K_0} = 1 + \frac{(\eta Z) K_0^{-1} (\eta Z)^*}{\sigma} \geq 1.} \quad (\text{PDS14})$$

For proof, block inversion of $\overline{H}_1$ gives for every (x_0, x_{top})

$$\begin{bmatrix} x_0 \\ x_{\text{top}} \end{bmatrix}^* \overline{H}_1^{-1} \begin{bmatrix} x_0 \\ x_{\text{top}} \end{bmatrix} = x_0^* \overline{H}_0^{-1} x_0 + \frac{|x_{\text{top}} - b^T \overline{H}_0^{-1} x_0|^2}{\sigma}. \quad (\text{PDS15})$$

Insert PDS11 and then $x_0 = Zz$. This proves the matrix equality. The determinant identity follows by taking the determinant of $I + K_0^{-1/2}(\eta Z)^*(\eta Z)K_0^{-1/2}/\sigma$; this rank-at-most-one operator has its only possible nonzero eigenvalue $(\eta Z)K_0^{-1}(\eta Z)^*/\sigma$. The frame Z has not been changed: replacing it by ZT changes both determinants by the same $|\det T|^2$. PDS14 also proves that the actual truncation $\mathcal{P}_1 \rightarrow \mathcal{Z}$ is a contraction for the displayed native metrics, and locates its entire metric defect in this one explicit row.

1.5 An explicit coefficient-product upper estimate

Set $D = D_1 = Q - v$ and keep the literal original polynomial

$$r_*(S') = \mathcal{T}_A(\chi_k R_{i_*})(S') = \mathcal{T}_{B_{i_*}} \mathcal{T}_A \chi_j(S'), \quad \lambda_* = \text{lc } r_* = b_* \mu_v \binom{Q}{v} \neq 0. \quad (\text{PDS16})$$

There is an exact finite upper bound

$$1 \leq \frac{\det K_1}{\det K_0} \leq \frac{\|r_*\|_{s,c}^2}{|b_*|^2 \sigma} \leq \frac{\|r_*\|_{s,c}^2}{|\lambda_*|^2 h_{D,s}}. \quad (\text{PDS17})$$

To prove the first upper inequality, the orthogonal projector onto $\text{im}(\overline{H}_0^{-1/2} Z)$ gives $Z K_0^{-1} Z^* \leq \overline{H}_0$. Thus the numerator in PDS14 is at most $\eta \overline{H}_0 \eta^*$. Let $v_* = u_*/b_* + H_0^{-1} b$. Then $\eta = -v_*^\top$, so this upper bound is $v_*^* H_0 v_*$. The orthogonal projection of r_* onto the old relation span has coefficient vector $b_* v_*$, whereas its orthogonal remainder is b_* times the remainder of r_{ℓ_1} . Consequently

$$\|r_*\|_{s,c}^2 = |b_*|^2 (v_*^* H_0 v_* + \sigma). \quad (\text{PDS18})$$

This proves the first inequality of PDS17. Every old relation has degree at most $D - 1$. The coefficient of ϕ_D in r_{ℓ_1} is $i^D \mu_v \binom{Q}{v} \sqrt{h_{D,s}}$; this uses its actual leading coefficient and the phase $i^{-D}/\sqrt{h_{D,s}}$ in PDS6. This coefficient remains in its orthogonal remainder. Hence $\sigma \geq |\mu_v \binom{Q}{v}|^2 h_{D,s}$, proving the second inequality of PDS17 with its original scalar intact.

For a completely finite coefficient bound define

$$L = \max\{1, |c| + 2(D + 1 + s/2)\}, \quad \Theta_{a,b,r,t;u,w} = \beta_{8,a,b} + \beta_{4,r,t} - \beta_{j,u,w},$$

$$P_* = \sum_{a,b=0}^8 \sum_{r,t=0}^4 |a_{ab} b_{i_*,r,t}| \prod_{u,w=0}^j (L + |\Theta_{a,b,r,t;u,w}|), \quad U_* = \frac{M_s P_*^2}{|b_* \mu_v \binom{Q}{v}|^2 h_{D,s}}. \quad (\text{PDS19})$$

The L in this subsection is a numerical bound and is not the jet length denoted L in LCA2. Then

$$0 \leq \log \frac{\det K_1}{\det K_0} \leq \log U_*. \quad (\text{PDS20})$$

Here is a direct proof that PDS19 bounds the original Gamma norm. In the orthonormal polynomial basis in y , multiplication by y has raising coefficient $\sqrt{(n+1)(n+s/2)}$ and lowering coefficient $\sqrt{n(n-1+s/2)}$, by PDS6 and $h_{n+1,s}/h_{n,s} = (n+1)(n+s/2)$. On polynomials of degree at most $D-1$ both weighted shifts have norm at most $D+1+s/2$; their sum has norm at most $2(D+1+s/2)$. Thus multiplication by $S' = c + iy$, through degree D , has norm at most L . Starting with the constant polynomial, whose squared norm is M_s , proves

$$\|p\|_{s,c} \leq \sqrt{M_s} \sum_{m=0}^D |[S'^m]p| L^m \quad (\deg p \leq D). \quad (\text{PDS21})$$

The exact expansion of PDS16 is

$$r_*(S') = \sum_{a,b=0}^8 \sum_{r,t=0}^4 a_{ab} b_{i_*,r,t} \prod_{u,w=0}^j (S' + \Theta_{a,b,r,t;u,w}). \quad (\text{PDS22})$$

The weighted coefficient ℓ^1 norm is submultiplicative: for two polynomials its coefficient convolution and the triangle inequality give $\sum_m |(fg)_m| L^m \leq (\sum_m |f_m| L^m)(\sum_m |g_m| L^m)$. Applying this identity to every factor in PDS22 bounds the weighted coefficient sum of r_* by P_* . Its coefficients above degree $D = Q - v$ vanish by the exact moment identities in PDS5; bounding the full degree- Q expansion before those cancellations is still an upper bound for the unchanged degree- D polynomial. PDS21 now gives $\|r_*\|_{s,c}^2 \leq M_s P_*^2$, and PDS17 proves PDS20. In particular $U_* \geq 1$ is a consequence, rather than an extra assumption on the original coefficients.

There is also an explicit uniform growth bound in either original source-order family. Define the fixed positive numbers

$$\begin{aligned} R_* &= \sqrt{(1/2 + \delta)^2 + \gamma^2}, \quad A_{\text{abs}} = \sum_{a,b=0}^8 |a_{ab}|, \quad B_{\text{abs}} = \sum_{r,t=0}^4 |b_{i_*,r,t}|, \\ C_* &= 6 + 17R_*, \quad C_{\text{abs}} = \max \left\{ 1, \frac{2(A_{\text{abs}}B_{\text{abs}})^2}{|b_*\mu_v|^2} \right\}. \end{aligned} \quad (\text{PDS23})$$

The original nodes satisfy $|\text{Re } \beta_{n,a,b}| \leq n(1/2 + \delta)$ and $|\text{Im } \beta_{n,a,b}| \leq n\gamma$. Both bounds are attained together at $(a, b) = (n, 0)$ and (n, n) ; thus $\max_{a,b} |\beta_{n,a,b}| = nR_*$, and therefore $|\Theta| \leq (k + 16)R_*$. For $s = 1$ or $s = k$, $D \leq Q$, $k \leq Q$, and $|c| \leq k/2 + 4$, so $L \leq 2Q + 3k/2 + 6 \leq 6(Q + 1)$. Also $k + 16 \leq 17(Q + 1)$. Every factor in PDS19 is consequently at most $C_*(Q + 1)$, giving $P_* \leq A_{\text{abs}}B_{\text{abs}}[C_*(Q + 1)]^Q$. The full factorial denominator gives a sharper bound than dropping its growth. For $s \geq 1$,

$$D!(s/2)_D \geq D!(1/2)_D = \frac{(2D)!}{4^D} \geq \left(\frac{D}{\exp(1)} \right)^{2D}.$$

The product identity follows by separating the even and odd factors of $(2D)!$. The last inequality follows from $\sum_{r=1}^{2D} \log r \geq \int_1^{2D} \log x \, dx = 2D \log(2D) - 2D + 1$. Since $D = Q - v$, $Q \geq 196$ and $v \leq 80$ imply $Q + 1 \leq 2D$, the exact PDS19 bound therefore satisfies

$$\begin{aligned} U_* &\leq \frac{(A_{\text{abs}}B_{\text{abs}})^2}{|b_*\mu_v|^2} C_*^{2Q} (Q + 1)^{2Q} \left(\frac{\exp(1)}{D} \right)^{2D} \\ &\leq C_{\text{abs}}(Q + 1)^{2v} (2 \exp(1) C_*)^{2Q}. \end{aligned}$$

Here $\binom{Q}{v} \geq 1$ was used only in an upper bound; the full binomial, mass and factorial denominator remain in PDS19. For the second inequality split $(Q + 1)^{2Q}$ into its $2v$ and $2D$ powers and use $Q + 1 \leq 2D$, $D \leq Q$ and $C_* \geq 6$. Consequently the proved finite bound is

$$0 \leq \log \frac{\det K_1}{\det K_0} \leq \log U_* \leq \log C_{\text{abs}} + 2v \log(Q + 1) + 2Q \log(2 \exp(1) C_*) = O_{\text{actual}}(Q) = o(kq_k).$$

(PDS24)

The last limit uses $Q/q_k \rightarrow 1$, fixed v , and $1/k \rightarrow 0$ for $k \equiv 1 \pmod{4}$. The constants are fixed by the actual quartet and period, independent of k and of the choice between the two original source-order families. The sharp finite PDS19 retains the full mass, phase-dependent nodes, factorials, conductor coefficients and leading scalar; PDS24 proves the scale of its upper bound without assuming an inverse estimate.

The maps PDS11 and PDS14 now give the primitive part of the low-cutoff source comparison inside the original LCA connecting sequence. The full cohomology additionally has the retained $\ker U_e$ constituent from LCC. Passing these source metrics through the actual connecting map requires the original target quotient Gram; PDS9 specifies the exact source dual to be used in that calculation, and PDS20–24 control its adjacent source correction. None of these equalities identifies a later target coefficient flag with a common compatible source at all cutoffs.

2 The original connecting image and its full native target determinant

Keep the complete CNC, LCC and LCA data, including the nonreduced A^{hom} , every original coefficient and the exact $E_A(z) = \sum a_{ab} e^{\beta_{s,a,b} z}$. Put

$$\begin{aligned} k &= 4l + 1 \geq 9, \quad j = k + 4, \quad q_a = (a + 1)^2, \quad q'_a = (a - 7)^2, \\ e &\in \{0, 1\}, \quad D_e = q_j - v - 1 + e, \quad L_e = D_e + 1, \quad 0 \leq v \leq 80, \quad \mu_v = E_A^{(v)}(0) \neq 0, \\ (s_0, s_1) &= (1, 1) \text{ or } (k, j), \quad c'_a = a/2 - 4, \quad m_a = q_a - q'_a - v. \end{aligned} \quad (\text{CIT1})$$

The two source orders in this display remain paired throughout; write $s_k = s_0$ and $s_j = s_1$ when a degree subscript is used. All finite functionals are complex-linear. At a fixed D_e let $\mathcal{H}_0, \mathcal{H}_1$ be their original Gamma polynomial duals in degrees k, j , with full orthonormal frames

$$\phi_{r,s,c'}(S') = \frac{p_{r,s}((S' - c')/i)}{\sqrt{(2\pi)^{s/2} r! (s/2)_r}}, \quad 0 \leq r \leq D_e. \quad (\text{CIT2})$$

Use the exact CNC10 matrices $T_0 = T_{k,D_e}$, $T_1 = T_{j,D_e}$ for its original analytic numerators divided by E_A , and V_0, V_1 for every original lower evaluation. Thus $[V_0, T_0]$ has its original full column order.

2.1 The literal primitive domain and target coefficient matrix

The four multiplier matrices are completely specified without an unspecified evaluator. Let $\mathcal{B}_a$ be the original monomial-to-Gamma matrix defined by $S'^r = \sum_n (\mathcal{B}_a)_{nr} \phi_{n,s_a,c'_a}$ and set

$$U_a = \text{diag}(1/r!)_{r=0}^{D_e} \mathcal{B}_a^\top, \quad (\text{Mult}_{L_e}(\mathcal{B}_i))_{rt} = \begin{cases} B_i^{(r-t)}(0)/(r-t)!, & r \geq t, \\ 0, & r < t, \end{cases} \quad (\text{CIT3})$$

$$\mathsf{T}_i = U_j^{-1} \text{Mult}_{L_e}(\mathcal{B}_i) U_k, \quad \mathsf{T} = \text{stack}_{i=1}^4 \mathsf{T}_i, \quad \mathcal{V} = \text{diag}(V_1, 4), \quad \mathcal{T}_1 = \text{diag}(T_1, 4).$$

Here $\text{diag}(P, 4)$ means four copies. The map U_a sends the actual native functional values to Taylor coefficients: ordinary derivatives are $\mathcal{B}_a^\top w$. This proves CIT3 and preserves every factorial and the original leading phase i^r inside $\mathcal{B}_a$. Its full inverse tuple is the analogous multiplication by the original entire D_i ; $\sum D_i \mathcal{B}_i = 1$ gives $\mathsf{ST} = I$.

Define the raw source analytic space

$$\mathcal{S}_k^{\text{raw}} = \text{im}[V_0, T_0] \subset \mathcal{H}_0, \quad \mathcal{L}_0 = \text{im } V_0, \quad \mathcal{L}_1 = \text{im } \mathcal{V}. \quad (\text{CIT4})$$

It is exactly the set of native functionals of N/E_A for all original upper numerators N vanishing to order v . Indeed T_0 gives the original DNE quotient frame and V_0 gives exactly its allowed numerator changes by conductor multiples. The columns are independent: if $T_0 x + V_0 \eta = 0$, the CNC13 injectivity at this cutoff gives $x = 0$, and the lower Vandermonde gives $\eta = 0$. Thus $\dim \mathcal{S}_k^{\text{raw}} = q_k - v$.

For the upper endpoint set

$$P_A = \mathcal{T}_A \chi_j, \quad (\mathcal{T}_A p)(S') = \sum_{a,b=0}^8 a_{ab} p(S' + \beta_{8,a,b}),$$

$$R_i(S') = \mathcal{T}_{B_i} P_A(S'), \quad (\mathcal{T}_{B_i} p)(S') = \sum_{a,b=0}^4 b_{i,a,b} p(S' + \beta_{4,a,b}), \quad (\text{CIT5})$$

$$\mathsf{R}_1 = \text{stack}_{i=1}^4 (\mathcal{B}_k \text{coeff}_{S'} R_i)^\top, \quad \mathsf{R}_0 : \mathcal{H}_0 \longrightarrow 0.$$

All coefficient columns are padded through D_1 . Here $\deg P_A = q_j - v = D_1$ and $\text{lc } P_A = \mu_v \binom{q_j}{v}$. The complete LCC9–12 proof shows that R_1 has rank four: the four original L_i^4 are independent on the rank-four Segre quadric, and translation of this polynomial is injective on the 25 distinct original shifts. The native row in CIT5 is an ordinary transpose because it evaluates the complex-linear functional on R_i .

The complete primitive domain is

$$\mathcal{Y}_e^{\text{raw}} = \ker \mathsf{R}_e, \quad \mathcal{S}_k^{\text{raw}} \subset \mathcal{Y}_e^{\text{raw}}, \quad r_e = \dim(\mathcal{Y}_e^{\text{raw}} / \mathcal{S}_k^{\text{raw}}) = 8k + 24 - 3e. \quad (\text{CIT6})$$

For $e = 0$ it is all of $\mathcal{H}_0$. To prove the target condition, at that endpoint the raw analytic image in each $\mathcal{H}_1$ has dimension $q_j - v = L_0$ and fills it. At $e = 1$ its only missing coordinate is $\Lambda_F(P_A)$ by LCC10. Therefore $\mathcal{T}h$ belongs to the fourfold raw target analytic image precisely when $\mathsf{R}_1 h = 0$. The exact numerator cochain square kills $\mathcal{S}_k^{\text{raw}}$ in those rows. Subtracting $\dim \mathcal{S}_k^{\text{raw}} = q_k - v$ from $\dim \ker \mathsf{R}_e = L_e - 4e$ proves CIT6.

Choose a full primitive lift matrix P_e so that $[V_0, T_0, P_e]$ is a frame of $\mathcal{Y}_e^{\text{raw}}$. This is an exact computable choice: first express CIT4–5 in ordinary Taylor coordinates, take the first nonzero pivot minors in their

original order to obtain the kernel and a complement, then transport those complement columns by U_k^{-1} . It depends on the actual coefficients and imposes no metric conditioning estimate. The chosen Taylor columns are fixed; the original norm remains CIT2.

In the original target coefficient coordinates define

$$\begin{aligned} P_{\mathcal{L}_1} &= \mathcal{V}(\mathcal{V}^* \mathcal{V})^{-1} \mathcal{V}^*, \quad K_1 = (I - P_{\mathcal{L}_1}) \mathcal{T}_1, \quad \mathcal{D}_e = K_1^* K_1 = \text{diag}(D_{j,D_e}^{(s_1)}, 4), \\ F_e &= \mathcal{D}_e^{-1} K_1^* (I - P_{\mathcal{L}_1}) \mathbb{T} P_e, \quad B_{\text{cof}} = \text{stack}_{i=1}^4 M_{i,k}. \end{aligned} \quad (\text{CIT7})$$

The inverses exist by the full lower Vandermonde and CNC13. By CIT6 the projected columns on the right belong to $\text{im } K_1$, so the displayed left inverse gives their unique original coefficient coordinates. Consequently

$$K_1 F_e = (I - P_{\mathcal{L}_1}) \mathbb{T} P_e, \quad K_1 B_{\text{cof}} = (I - P_{\mathcal{L}_1}) \mathbb{T} T_0. \quad (\text{CIT8})$$

These are literal original finite matrices. In particular the $M_{i,k}$ retain the exact KRC scalar-bearing conjugation CNC12.

2.2 Complete connecting kernel and attained target norm

We prove the full kernel of the connecting construction. The exact finite intersection needed here is

$$\text{im } \mathbb{T} \cap \mathcal{L}_1 = \mathbb{T} \mathcal{L}_0. \quad (\text{CIT9})$$

For a direct proof, write $\mathbb{T}h = \mathcal{V}\eta$. Apply the six original compatibility rows. Their coefficient square is $\mathcal{R}\mathcal{V} = \mathcal{V}_2 \mathcal{K}$, so $\mathcal{V}_2 \mathcal{K}\eta = 0$. At the present cutoffs $D_e + 1 - q_k = 8k + 24 - v + e \geq 16$; hence the original degree- k evaluation $\mathcal{V}_2$ is injective. The complete GKC kernel gives $\eta = \mathcal{P}\xi$. The lower root-addition square says $\mathcal{V}\mathcal{P} = \mathbb{T}V_0$, whence $\mathbb{T}h = \mathbb{T}V_0\xi$. Since $\mathbb{S}\mathbb{T} = I$, $h = V_0\xi$. The reverse inclusion is exactly that same square. This proves CIT9 without a transversality or inverse-norm assumption.

For $h \in \mathcal{Y}_e^{\text{raw}}$ let $c(h)$ be the coefficient column defined as in CIT7. The native exterior differential of $\mathbb{T}h$ is zero, since $B_i B_t = B_t B_i$. Replacing its representative by lower evaluations changes its next differential by lower evaluations, by root addition. Therefore $c(h)$ is a cycle in the actual LCC image complex. The complete map is

$$\begin{aligned} \partial_e : \mathcal{Y}_e^{\text{raw}} / \mathcal{S}_k^{\text{raw}} &\longrightarrow H^1(\mathcal{I}_e), \quad [h] \longmapsto [c(h)], \\ \text{im } \partial_e &= \mathcal{C}_e, \quad \mathcal{C}_e = \mathcal{S}_e / \mathcal{B}, \quad \mathcal{S}_e = \text{im}[B_{\text{cof}}, F_e], \quad \mathcal{B} = \text{im } B_{\text{cof}}. \end{aligned} \quad (\text{CIT10})$$

If $h = T_0 x + V_0 \xi$, its target coefficients are the boundary $B_{\text{cof}} x$ by CIT8. Conversely, if $c(h) = B_{\text{cof}} x$, then $\mathbb{T}(h - T_0 x) \in \mathcal{L}_1$ and CIT9 gives $h - T_0 x \in \mathcal{L}_0$. Thus the kernel before quotient is exactly $\mathcal{S}_k^{\text{raw}}$. This proves injectivity and the stated presentation; $[B_{\text{cof}}, F_e]$ is independent. It is exactly the positive connecting image of $H^0(\mathcal{R}_e) = \ker \mathfrak{t}_e$ in LCC13, since its construction lifts the original degree-zero quotient cycle and takes its differential with the same positive sign. No splitting of the remaining $\ker U_e$ has been introduced.

Its full cohomology restriction Gram in the P_e image coordinates is

$$\begin{aligned} H_{\text{conn},e} &= F_e^* \mathcal{D}_e F_e \\ &\quad - F_e^* \mathcal{D}_e B_{\text{cof}} (B_{\text{cof}}^* \mathcal{D}_e B_{\text{cof}})^{-1} B_{\text{cof}}^* \mathcal{D}_e F_e. \end{aligned} \quad (\text{CIT11})$$

The exact least norm over every original boundary gives this Schur complement. All original lower coefficients were already minimized simultaneously in $\mathcal{D}_e$. Equivalently set

$$Z_e^{\text{raw}} = [\mathcal{V}, \mathbb{T}T_0], \quad H_{\text{conn},e} = (\mathbb{T}P_e)^* (I - P_{Z_e^{\text{raw}}}) \mathbb{T}P_e, \quad P_Z = Z(Z^* Z)^{-1} Z^*. \quad (\text{CIT12})$$

The columns of Z_e^{raw} are independent by CIT9 and the independence in CIT4. The same argument with the added P_e proves independence of $[Z_e^{\text{raw}}, \mathbb{T}P_e]$. Indeed the complete minimization in CIT11 allows exactly $\mathbb{T}T_0 x + \mathcal{V}\eta$. This proves CIT12 and positivity. It also gives the full original raw determinant

$$\det H_{\text{conn},e} = \frac{\det([Z_e^{\text{raw}}, \mathbb{T}P_e]^* [Z_e^{\text{raw}}, \mathbb{T}P_e])}{\det((Z_e^{\text{raw}})^* Z_e^{\text{raw}})}. \quad (\text{CIT13})$$

These are the same fixed combined frames, so their ordinary Schur determinant introduces no missing coordinate factor.

2.3 The original source-to-image determinant at its low cutoff

Give the primitive source quotient its original source metric and its correlated image metric, respectively:

$$H_{\text{src},e} = P_e^*(I - P_{[V_0, T_0]})P_e, \quad H_{\text{cor},e} = (\mathbb{T}P_e)^*(I - P_{\mathbb{T}[V_0, T_0]})\mathbb{T}P_e. \quad (\text{CIT14})$$

These are two explicit attained minima. The original NBG $\|\mathbb{T}\| \leq K_+$ and $\|\mathbb{S}\| \leq K_-$ with $\mathbb{S}\mathbb{T} = I$ imply

$$K_-^{-2}H_{\text{src},e} \preceq H_{\text{cor},e} \preceq K_+^2H_{\text{src},e}. \quad (\text{CIT15})$$

For each fixed primitive value, apply the two norm inequalities to every representative in $P_e b + \mathcal{S}_k^{\text{raw}}$ and take the minima. This proves both directions on the exact same fibre. The finite constants are the complete NBG12 ones at base k and cutoff D_e , including source order $(1, 1)$ or (k, j) and center shift $(\beta_{4,a,b} - 2)/i$.

There is also an exact independent-gauge quotient identity. In the original source quotient put

$$X_k = \mathcal{S}_k^{\text{raw}}/\mathcal{L}_0, \quad Y_e = \mathcal{Y}_e^{\text{raw}}/\mathcal{L}_0, \quad X_k \subset Y_e.$$

Their correlated target images are the original NKC9 images. The original independent-gauge form restricted to Y_e and then quotiented by X_k allows precisely the total fibre $\mathbb{T}T_0x + \mathcal{L}_1$; its correlated quotient allows $\mathbb{T}T_0x + \mathbb{T}\mathcal{L}_0$. These are CIT12 and CIT14. Hence the literal PCJ7 source quotient gives

$$\frac{\det H_{\text{conn},e}}{\det H_{\text{cor},e}} = \det \Omega_{Y_e/X_k, D_e}. \quad (\text{CIT16})$$

If R transports the original primitive lifts to an adapted correlated quotient frame, both Grams are changed by $R^*(\cdot)R$, and their two factors $|\det R|^2$ cancel in CIT16. This is the exact coordinate map behind that determinant equality.

Use the actual compatibility-plus-complement expressions χ_{X_k, D_e} and χ_{Y_e, D_e} of PCJ1. The complete shared-update proof PCJ13–16 gives

$$\begin{aligned} \log \det H_{\text{conn},e} - \log \det H_{\text{src},e} &= -(\chi_{X_k, D_e} - \chi_{Y_e, D_e}) + \epsilon_e, \\ -2r_e \log \max(1, K_-) - r_e \log c_+ &\leq \epsilon_e \leq 2r_e \log \max(1, K_+) - r_e \log c_-. \end{aligned} \quad (\text{CIT17})$$

Indeed write the left side as the correlated/source logarithm plus $\log \det \Omega_{Y_e/X_k}$. The first is bounded by CIT15. PCJ16 writes the negative of the latter as $\chi_{X_k} - \chi_{Y_e} + \nu$, with $r_e \log c_- \leq \nu \leq r_e \log c_+$. Subtraction gives exactly CIT17. Here $c_- = \min(1, L_-^{-2})$ and $c_+ = \max(1, L_+^2)$ are the original NKC constants at this cutoff. Consequently this specified error is $O_{\text{actual}}(k \log q_k) = o(kq_k)$ in both original source pairs. At $e = 0$, Y_0 is the full ambient source quotient, so its source-complement projector vanishes and $\chi_{Y_0} = 0$ exactly. At $e = 1$ its four actual polynomial constraints remain; χ_{Y_1} is not set to zero.

CIT14–17 concern the defining low cutoff D_e . They do not assert that a primitive truncated germ has a common compatible extension to every later cutoff. The target coefficient columns F_e now are fixed, and their complete later metrics are calculated next without that assertion.

2.4 The fixed connecting flag at every original target row

For $q_j - 1 \leq N \leq 2q_j$ put $D = N - v$ and retain the fixed B_{cof}, F_e in the original target coefficients. Define

$$\begin{aligned} \mathcal{D}_N &= \text{diag}(D_{j, N-v}^{(s_1)}, 4), \quad Z_{B, N} = [\mathcal{V}_N, \mathcal{T}_{1, N} B_{\text{cof}}], \\ Z_{C, N} &= [Z_{B, N}, \mathcal{T}_{1, N} F_e], \quad H_{\text{conn}, e}(N) = \text{Gram}_{\mathcal{D}_N}(\mathcal{S}_e/\mathcal{B}), \\ \det H_{\text{conn}, e}(N) &= \det(Z_{C, N}^* Z_{C, N}) / \det(Z_{B, N}^* Z_{B, N}). \end{aligned} \quad (\text{CIT18})$$

The original DNE positivity and the independence in CIT10 make both raw frames full column rank at every one of these indices. Eliminating the complete lower frame and then B_{cof} proves the determinant identity. At the defining $N = q_j - 1 + e$, it is exactly CIT11–13 by CIT8.

Every entry is given by the original arithmetic recursion. If $\nu_{a,n} = \sum (Z_{\text{exp},j})_{uv,a} \beta_{j,u,v}^n$, then

$$f_{a,n} = \frac{\nu_{a,n+v} - \sum_{d=v+1}^{n+v} \binom{n+v}{d} \mu_d f_{a,n+v-d}}{\binom{n+v}{v} \mu_v},$$

$$b_{0,m} = \mathbf{1}_{m=0}, \quad b_{-1,m} = 0, \quad b_{n+1,m} = i^{-1} (b_{n,m-1} - c'_j b_{n,m}) - n(n-1 + s_1/2) b_{n-1,m}, \quad (\text{CIT19})$$

$$t_n = \left(\frac{\sum_m b_{n,m} f_{a,m}}{\sqrt{(2\pi)^{s_1/2} n! (s_1/2)_n}} \right)_a, \quad v_n = \left(\frac{\sum_m b_{n,m} \beta_{j-8,u,v}^m}{\sqrt{(2\pi)^{s_1/2} n! (s_1/2)_n}} \right)_{u,v}.$$

Arrays outside their polynomial range are zero. Differentiating $E_A F_a = N_a$ gives the first line; substituting the full original Gamma recurrence gives the next. Thus these are exactly the unchanged IRR7–9 rows, retaining all original moments and phases. The new four-row blocks in CIT18 are $z_{B,n} = [\text{diag}(v_n, 4), \text{diag}(t_n, 4) B_{\text{cof}}]$ and $z_{C,n} = [z_{B,n}, \text{diag}(t_n, 4) F_e]$. For $X = B, C$ define $G_{X,n} = Z_{X,n+v}^* Z_{X,n+v}$. In degree-major row order each raw Gram increases by $z^* z$. The fixed row permutation from copy-major order is unitary.

For $q_j - v \leq n \leq 2q_j - v$, the exact connecting-volume increment is

$$\log \frac{\det H_{\text{conn},e}(n+v)}{\det H_{\text{conn},e}(n+v-1)} = \log \det(I_4 + z_{C,n} G_{C,n-1}^{-1} z_{C,n}^*) - \log \det(I_4 + z_{B,n} G_{B,n-1}^{-1} z_{B,n}^*) \geq 0. \quad (\text{CIT20})$$

The determinant lemma proves the equality. For its sign, transform the old extra columns to their full least-norm section over $Z_{B,n+v-1}$. The larger row covariance is the smaller one plus $RQ^{-1}R^* \succeq 0$, by block inversion. This proves the inequality and retains the actual old quotient denominator. No independent rowwise gauge minimum is taken.

2.5 Its exact place in the common low invariant loss

Let $\mathbb{J} = \ker \mathbb{B}$ be the full original fourfold DNE invariant kernel. At each later N use the same actual target metric $\mathcal{D}_N$ and fixed flags

$$W = \mathcal{B} + \mathbb{J}, \quad K_{\text{conn},e} = \mathcal{S}_e \cap W = \mathcal{B} + (\mathcal{S}_e \cap \mathbb{J}),$$

$$T_{\text{conn},e} = \mathcal{S}_e + W = \mathcal{S}_e + \mathbb{J}, \quad (\text{CIT21})$$

$$\omega_{\text{conn},e}(N) = \frac{\det \text{Gram}_{\mathcal{D}_N}(T_{\text{conn},e}/W)}{\det \text{Gram}_{\mathcal{D}_N}(\mathcal{S}_e/K_{\text{conn},e})}.$$

The two Grams use the same quotient-value coordinates under $[z] \mapsto [\mathbb{B}z]$. Its original fibres are exactly the two spaces in the denominator and numerator, as in LCC20–23. The source is the corresponding quotient of the connecting cohomology image; its full kernel is $(\mathcal{S}_e \cap \mathbb{J})/(\mathcal{B} \cap \mathbb{J})$. Thus CIT21 is its actual further target loss, and is in $(0, 1]$.

The full low cycles contain $\mathcal{S}_e$, and LCC13 gives

$$\dim(\mathcal{Z}_e/\mathcal{S}_e) = \kappa_e \leq 3 \max(v - e, 0). \quad (\text{CIT22})$$

Both spaces are fixed original coefficient flags for the later metric comparisons. The difference of their invariant loss returns has the finite bound

$$|\mathcal{R} \log \omega_e - \mathcal{R} \log \omega_{\text{conn},e}| \leq 2\kappa_e \log A_{j,s_1}^{\text{end}} = O_{\text{actual}}(q_j \log j) = o(kq_k). \quad (\text{CIT23})$$

Here $\mathcal{R}f = -f(q_j - 1) - f(q_j) + f(2q_j - 1) + f(2q_j)$. For a complete verification, apply the actual inclusions $\mathcal{S}_e \subset \mathcal{Z}_e$ with their common W . The kernel/intersection quotient and sum quotient fit into the exact sequence $0 \rightarrow K_{\text{full}}/K_{\text{conn}} \rightarrow \mathcal{Z}_e/\mathcal{S}_e \rightarrow T_{\text{full}}/T_{\text{conn}} \rightarrow 0$ by the identity on representatives. Their two ranks sum to κ_e . The full adapted-Gram factorization LFC7–12 expresses the loss difference as the sum of those two quotient log determinants minus the $\mathcal{Z}_e/\mathcal{S}_e$ log determinant and a fixed frame constant. FRE's same native form sandwich puts the three monotone returns between zero and twice their ranks times $\log A_{j,s_1}^{\text{end}}$. Both sides of the resulting signed difference are bounded by $2\kappa_e \log A$. This proves CIT23, including $\kappa_e = 0$, with all frame constants canceled only after the original four signs.

There is a further exact relation to the ORIGINAL ITL source. Write $F_{\text{inv}} = \text{stack}_i(M_{i,k}J_0)$. Since $B_{\text{cof}} = \text{stack}_i M_{i,k}$,

$$\text{im } F_{\text{inv}} \subset \mathcal{B}, \quad \mathbb{J} \subset W_{\text{inv}} := \mathbb{J} + \text{im } F_{\text{inv}} \subset W \subset T_{\text{conn},e}. \quad (\text{CIT24})$$

These inclusions are literal original coefficient maps. The ITL source therefore represents boundaries in low cohomology, but its target Gram remains in the denominator $\text{Gram}(W)$ of the cohomology target quotient.

To exhibit every factor, let $I_R = F_{\text{inv}}Z_R$ be exactly the fixed ITL4 source-kernel complement and choose fixed columns R_b completing $[X_{\mathbb{J}}, I_R]$ to W . In a specified fixed full frame X_W , write

$$[X_{\mathbb{J}}, I_R, R_b] = X_W C_W, \quad Q_b(N) = \text{Gram}_{\mathcal{D}_N}(W/W_{\text{inv}}) \quad \text{in the lifts } R_b.$$

The first quotient of this chain has exactly ITL's target Gram $H_{\text{inv}}(N)$: its fibre is $I_R b + \mathbb{J}$. Its source pullback quotient is the full ITL $A_{\text{inv}}(N)$, and $\det H_{\text{inv}} = \det A_{\text{inv}} \omega_{\text{inv}}$. Sequential Schur complements consequently give

$$|\det C_W|^2 g_W(N) = g_{\mathbb{J}}(N) \det H_{\text{inv}}(N) \det Q_b(N), \quad g_X(N) = \det(X_X^* \mathcal{D}_N X_X). \quad (\text{CIT25})$$

All matrices in this formula are in their stated independent frames; $r_{\text{inv}} = 0$ gives the empty first quotient.

Finally retain the full LFC coordinate constant $c_{\text{conn},e}$ from the adapted frames for

$$(\mathcal{S}_e, K_{\text{conn},e}, T_{\text{conn},e}, W).$$

Its cross ratio and CIT25 give the exact original identity

$$\begin{aligned} \log \omega_{\text{conn},e} + \log \omega_{\text{inv}} &= \log g_{T_{\text{conn},e}} + \log g_{K_{\text{conn},e}} - \log g_{\mathcal{S}_e} - \log g_{\mathbb{J}} \\ &\quad - \log \det A_{\text{inv}} - \log \det Q_b + \log c_{\text{conn},e} + 2 \log |\det C_W|. \end{aligned} \quad (\text{CIT26})$$

This relates the two actual losses through their exact common boundary structure; it does not identify their source spaces. Taking the original four-return cancels the last two fixed coordinate constants. Equivalently ITL16's exact matched relative operator satisfies $\det H_{\text{inv}} = \det Q_{\text{inv}} \det \mathcal{R}_X$, where Q_{inv} is the full original source quotient $J_0^* D_{k,N-v}^{(s_0)} J_0 / E$. Substituting this equality instead in CIT25 replaces $(\omega_{\text{inv}}, A_{\text{inv}})$ in CIT26 by $(\det \mathcal{R}_X, Q_{\text{inv}})$, preserving the source-order pair $(1, 1)$ or (k, j) . It retains the original source mass.

Thus the surviving connecting target is given by the explicit matrix CIT7, its complete target minimum CIT11–13, the original low source-to-image determinant CIT17, and all later original row formulas CIT18–20. The fixed-rank remainder is controlled by CIT23. The remaining common target loss has the exact original ITL boundary relation CIT24–26, with its whole residual overlap, all absolute native Grams and both source measures still present.

3 Primitive-source transport inside the original connecting image

Keep the complete original PDS1–24 and CIT1–26 data at fixed $k = 4l + 1 \geq 9$, $j = k + 4$, $D_e = (k + 5)^2 - v - 1 + e$, and either original source-order pair $(s_0, s_1) = (1, 1)$ or (k, j) . All polynomial coefficients, period data, lower gauges and exterior signs remain those of those constructions. In this section $P_e, F_e, T_{a,D_e}, V_{a,D_e}$ are CIT's primitive lifts, connecting target columns, analytic frame and full lower frame. The polynomial coefficient matrix C_e and metric $G_e = (C_e^* C_e)^{-1}$ are PDS7. The boundary column matrix is the unchanged $B_{\text{cof}} = \text{stack}_i M_{i,k}$.

3.1 The exact source coordinate map and its target naturality

The coordinate maps between the two proved primitive presentations are

$$\Theta_e = C_e^T P_e, \quad \Theta_0 : \mathbb{C}^{d_0} \xrightarrow{\simeq} \mathbb{C}^{d_0}, \quad \Theta_1 : \mathbb{C}^{d_0-3} \xrightarrow{\simeq} \ker W^T, \quad d_0 = 8k + 24. \quad (\text{PDI1})$$

The inverse signs mean inverses onto the displayed images. Indeed PDS4 is the onto quotient by precisely CIT4's analytic source, and P_e is a full basis of that primitive quotient. The PDS8 four-row condition is CIT5's same polynomial differential. Thus both maps are bijections onto the asserted spaces. Equality of the exact minimum-norm fibres proves

$$H_{\text{src},e} = \Theta_e^* G_e \Theta_e. \quad (\text{PDI2})$$

For completeness a native vector w and $P_e a$ have the same PDS quotient coordinates exactly when $C_e^\top(w - P_e a) = 0$, equivalently $w - P_e a \in \text{im}[V_{k,D_e}, T_{k,D_e}]$ by PDS4. The minimum over this fibre is both sides of PDI2, by CIT14 and PDS7. This proves the metric identity as well as its linear map.

Let $J_k^{\text{tr}} : \mathcal{H}_{k,D_1} \rightarrow \mathcal{H}_{k,D_0}$ and $J_j^{\text{tr}} : \mathcal{H}_{j,D_1} \rightarrow \mathcal{H}_{j,D_0}$ be literal jet truncation, so they delete the last native functional coordinate; the source order and centre are fixed during this truncation. Let pr_0 delete the last polynomial quotient coordinate as in PDS11, and set

$$Z = \text{pr}_0 \Theta_1, \quad R = \Theta_0^{-1} Z, \quad \text{rank } R = d_0 - 3, \quad \Theta_1 = EZ. \quad (\text{PDI3})$$

The rank assertion and last equality follow from PDS11. The first d_0 relation polynomials are unchanged between cutoffs; therefore $C_0^\top J_k^{\text{tr}} P_1 = \text{pr}_0 \Theta_1 = \Theta_0 R$. Their difference is in the old analytic source. In its original independent frame there are unique matrices L and A such that

$$J_k^{\text{tr}} P_1 = P_0 R + V_{k,D_0} L + T_{k,D_0} A. \quad (\text{PDI4})$$

They are explicit finite matrices: with $U = [V_{k,D_0}, T_{k,D_0}]$ their exact block expression is

$$\begin{bmatrix} L \\ A \end{bmatrix} = (U^* U)^{-1} U^* (J_k^{\text{tr}} P_1 - P_0 R).$$

The actual connecting coefficient columns satisfy

$$\boxed{F_1 = F_0 R + B_{\text{cof}} A.} \quad (\text{PDI5})$$

To prove it, the original multiplier square commutes with truncation: $(\text{diag}(J_j^{\text{tr}}, 4))\mathsf{T}^{(1)} = \mathsf{T}^{(0)} J_k^{\text{tr}}$. This is the equality of the Taylor convolutions in CIT3 after discarding the last coefficient. Likewise truncation takes each original lower or analytic target column to the corresponding old column, because they are jets of the same entire or analytic germ. Apply these equalities to the target representative of F_1 in CIT8 and use PDI4. Its old target jet differs from that of $F_0 R + B_{\text{cof}} A$ by an allowed lower evaluation, using the root-addition square on $V_{k,D_0} L$. The original degree- j analytic coefficient map modulo lower evaluations is injective at D_0 by CNC13. Thus the coefficient columns themselves agree, proving PDI5. No inverse estimate is used to pass from equality of the jets to these coefficients.

In particular, in the single original target coefficient space,

$$\mathcal{S}_1 = \text{im}[B_{\text{cof}}, F_1] \subset \mathcal{S}_0 = \text{im}[B_{\text{cof}}, F_0], \quad \dim(\mathcal{S}_0/\mathcal{S}_1) = 3. \quad (\text{PDI6})$$

Independence of each displayed frame is proved in CIT10, and PDI3 gives the rank of the quotient inclusion. This proves the codimension and identifies the actual map from primitive truncation to the fixed connecting coefficient flags.

3.2 The two low metric increments in common coordinates

The old source restricted to this image and the upper source, in precisely the P_1 coordinates, have Grams

$$K_{\text{src},0} = R^* H_{\text{src},0} R = Z^* G_0 Z, \quad K_{\text{src},1} = H_{\text{src},1} = (EZ)^* G_1 (EZ). \quad (\text{PDI7})$$

PDS12–24 now apply to these actual connecting-source frames:

$$K_{\text{src},1} = K_{\text{src},0} + \frac{(\eta Z)^* (\eta Z)}{\sigma}, \quad 0 \leq \Delta_{\text{src}} := \log \frac{\det K_{\text{src},1}}{\det K_{\text{src},0}} \leq \log U_* = o(kq_k). \quad (\text{PDI8})$$

Every constant in U_* is the original PDS19 finite product; it has the fully proved explicit bound PDS24.

In the original target define

$$K_{\text{tgt},0} = R^* H_{\text{conn},0} R, \quad K_{\text{tgt},1} = H_{\text{conn},1}. \quad (\text{PDI9})$$

By PDI5 these are the metrics of the same fixed quotient $\mathcal{S}_1/\mathcal{B}$ at the two native target cutoffs, in the same F_1 value coordinates. Adding the boundary $B_{\text{cof}} A$ does not change its minimum over that boundary. This

proves the first identity in PDI9 as a quotient statement, including its frame, rather than presuming equality of two arbitrary representatives.

Here is its complete four-row update. At D_0 set

$$\begin{aligned} N_B &= [\text{diag}(V_{j,D_0}, 4), \text{diag}(T_{j,D_0}, 4)B_{\text{cof}}], & N_F &= \text{diag}(T_{j,D_0}, 4)F_1, \\ G_B &= N_B^* N_B, & G_{BF} &= N_B^* N_F, & K_{\text{tgt},0} &= N_F^* N_F - G_{BF}^* G_B^{-1} G_{BF}. \end{aligned} \quad (\text{PDI10})$$

Both original lower and boundary frames are kept in N_B . Let z_B, z_F be the next degree's four rows of these very frames, computed from CIT19, and define

$$S_B = I_4 + z_B G_B^{-1} z_B^*, \quad r_B = z_F - z_B G_B^{-1} G_{BF}. \quad (\text{PDI11})$$

Their exact update is

$$\begin{aligned} K_{\text{tgt},1} &= K_{\text{tgt},0} + r_B^* S_B^{-1} r_B, \\ \Delta_{\text{tgt}} &:= \log \frac{\det K_{\text{tgt},1}}{\det K_{\text{tgt},0}} = \log \det \left(I_4 + S_B^{-1/2} r_B K_{\text{tgt},0}^{-1} r_B^* S_B^{-1/2} \right) \geq 0. \end{aligned} \quad (\text{PDI12})$$

For proof replace N_F by its old orthogonal residual $N_F - N_B G_B^{-1} G_{BF}$. In this unchanged quotient the old joint Gram is $\text{diag}(G_B, K_{\text{tgt},0})$; the added rows are $[z_B, r_B]$. The new Schur complement is $K_{\text{tgt},0} + r_B^* [I - z_B (G_B + z_B^* z_B)^{-1} z_B^*] r_B$. Multiplying by $I + z_B G_B^{-1} z_B^*$ proves that the bracket is S_B^{-1} . Taking the determinant gives PDI12. The column operation has identity diagonal blocks, so its determinant is one and it leaves no missing frame factor.

There is a finite bound using the complete original native next-row budget. Write $D_0 = D_{j,D_0}^{(s_1)}$ for the single-copy original coefficient metric, $T = T_{j,D_0}$, $V = V_{j,D_0}$, and let t, v_{row} be their next original rows. Set

$$\begin{aligned} A_V &= V^* V, \quad \zeta = v_{\text{row}} A_V^{-1} v_{\text{row}}^*, \quad \rho = t - v_{\text{row}} A_V^{-1} V^* T, \\ \beta_{\text{next}} &= 4 \log \left(1 + \frac{\rho D_0^{-1} \rho^*}{1 + \zeta} \right), \quad 0 \leq \Delta_{\text{tgt}} \leq \beta_{\text{next}} \leq 4 \log A_{j,s_1}^{\text{end}} = o(kq_k). \end{aligned} \quad (\text{PDI13})$$

The exact lower-gauge Schur update is $D_1 = D_0 + \rho^* \rho / (1 + \zeta)$, as in IRR and LFC. Thus the full four-copy ambient log-determinant increment is β_{next} . To prove the middle bound without multiplying by a quotient dimension, put the old ambient form into its isometric coordinates and write the new one as e^B with $B \succeq 0$. For any fixed subspace with full frame Y , differentiation along e^{tB} gives

$$\frac{d}{dt} \log \det(Y^* e^{tB} Y) = \text{Tr}(B P_Y(t)), \quad P_Y(t) = e^{tB/2} Y (Y^* e^{tB} Y)^{-1} Y^* e^{tB/2}.$$

$P_Y(t)$ is the orthogonal projector onto $e^{tB/2} Y$. For nested fixed spaces its two projectors are ordered, so the derivative of their quotient log determinant is between zero and $\text{Tr } B$. Integrating proves that the increment of $\mathcal{S}_1/\mathcal{B}$ is at most the full ambient increment, which is β_{next} . Finally FRE's original endpoint form bound implies that each of the four identical single-copy rank-one relative eigenvalues is at most A_{j,s_1}^{end} , proving the last inequality. Its already proved growth is $\log A_{j,s_1}^{\text{end}} = O_{\text{actual}}(q_j \log j) = o(kq_k)$. The isometric coordinates in this proof establish a comparison of the unchanged forms; the formulas PDI10–13 remain in their original frames.

The exact adjacent source-to-image determinant correction is therefore the difference of the explicit four-row and one-row log determinants PDI12 and PDI8:

$$\begin{aligned} \Delta_{\text{image/source}} &:= \log \frac{\det K_{\text{tgt},1}}{\det K_{\text{src},1}} - \log \frac{\det K_{\text{tgt},0}}{\det K_{\text{src},0}} = \Delta_{\text{tgt}} - \Delta_{\text{src}}, \\ -\log U_* &\leq \Delta_{\text{image/source}} \leq \beta_{\text{next}}, \quad \Delta_{\text{image/source}} = o(kq_k). \end{aligned} \quad (\text{PDI14})$$

The lower determinant uses the proved common primitive subspace R ; its three-dimensional complement has not been silently discarded from the full old cohomology.

3.3 The two fixed connecting invariant losses

At every later original target cutoff keep the literal flags of CIT21 and the same $W = \mathcal{B} + \mathbb{J}$. By PDI6 their nesting has codimension exactly three. The identity on representatives gives the exact sequence

$$0 \longrightarrow K_0/K_1 \longrightarrow \mathcal{S}_0/\mathcal{S}_1 \longrightarrow T_0/T_1 \longrightarrow 0, \quad K_e = \mathcal{S}_e \cap W, \quad T_e = \mathcal{S}_e + W. \quad (\text{PDI15})$$

Surjectivity follows from the definition of the sum; its kernel consists of the class of $x \in \mathcal{S}_0$ with $x \in \mathcal{S}_1 + W$, hence of $K_0 + \mathcal{S}_1$ modulo $\mathcal{S}_1$. Its kernel presentation is K_0/K_1 . Thus the two end ranks sum to three. The complete adapted Gram factorization LFC7–12 expresses the difference of the two loss logarithms as $\log \det \text{Gram}(T_0/T_1) + \log \det \text{Gram}(K_0/K_1) - \log \det \text{Gram}(\mathcal{S}_0/\mathcal{S}_1)$ plus a fixed coordinate constant. Under the original four-return $\mathcal{R}f = -f(q_j - 1) - f(q_j) + f(2q_j - 1) + f(2q_j)$ that constant cancels. FRE bounds each monotone quotient return between zero and twice its rank times $\log A_{j,s_1}^{\text{end}}$. The positive two ranks sum to three and the negative rank is three; therefore

$$|\mathcal{R} \log \omega_{\text{conn},0} - \mathcal{R} \log \omega_{\text{conn},1}| \leq 6 \log A_{j,s_1}^{\text{end}} = o(kq_k). \quad (\text{PDI16})$$

This is the complete primitive-image refinement of the full-cohomology comparison in LFC. The retained $\ker U_e$ constituents of the full LCC sequence are still present, with their explicit CIT23 bounds. The common long connecting loss continues to be the exact target expression CIT21 and its ITL boundary factorization CIT24–26. The original low source transport now feeds that target calculation through the explicit coefficient identity PDI5, the two native increments PDI12–14, and the proved fixed-flag comparison PDI16.